

Year 7 · Chapter 8 · Area and volume: answers (teacher copy)

8.01 Units of measurement (booklet p. 3)

Examples (model these live)

E1 Change 3.5 km to metres.
km to m: $\times 1000$ $\cdot = 3\,500$ m

E2 Change 450 g to kilograms.
g to kg: $\div 1000$ $\cdot = 0.45$ kg

E3 Add 2 m and 35 cm. Give the answer in centimetres.
 $2\text{ m} = 200\text{ cm}$ $\cdot 200 + 35 = 235\text{ cm}$

Questions (40 on today's skill, 17 from earlier lessons)

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|--|---|---|---|
| 1 23 m | 20 $605\text{ cm} = 6.05\text{ m} \cdot 5 + 6.05 = 11.05\text{ m}$ | 32 $6\text{ L} = 6000\text{ mL} \cdot 6000 - 311 = 5689\text{ mL}$ | 44 $102.6\text{ cm}, 105\text{ cm}, 108\text{ cm} \cdot 1\,026\text{ mm}, 1.05\text{ m}, 108\text{ cm}$ |
| 2 51 000 mL | 21 $1\text{ km} = 1\,000\text{ m} \cdot 3.45 \times 1\,000 = 3\,450\text{ m}$ ^{7.04} | 33 $328\text{ cm} = 3.28\text{ m} \cdot 2 + 3.28 = 5.28\text{ m}$ | 45 0.1 ^{7.04} |
| 3 240 minutes | 22 $4 \times 60 = 240 \cdot 240 + 17 = 257\text{ min}$ | 34 0.090, 0.200, 0.699, 0.900 $\cdot 0.9, 0.699, 0.2, 0.09$ ^{7.01} | 46 $7 \times 365 = 2555\text{ g} \cdot = 2.555\text{ kg}$ |
| 4 21 000 m | 23 km to m: $\times 1\,000 \cdot 36 \times 1\,000 = 36\,000\text{ m}$ | 35 $2 \times 60 = 120 \cdot 120 + 13 = 133\text{ min}$ | 47 $1\text{ L} = 1\,000\text{ mL} \cdot 0.28 \times 1\,000 = 280\text{ mL}$ ^{7.04} |
| 5 38 000 g | 24 $p = 1$ ^{5.12} | 36 $6\,470$ ^{7.04} | 48 $116\text{ cm}, 112.8\text{ cm}, 121\text{ cm} \cdot 1\,128\text{ mm}, 1.16\text{ m}, 121\text{ cm}$ |
| 6 120 seconds | 25 $737\text{ cm} = 7.37\text{ m} \cdot 7 + 7.37 = 14.37\text{ m}$ | 37 m to cm: $\times 100 \cdot 33 \times 100 = 3\,300\text{ cm}$ | 49 $11 \times 243 = 2673\text{ g} \cdot = 2.673\text{ kg}$ |
| 7 32 m | 26 $2 \times 60 = 120 \cdot 120 + 49 = 169\text{ min}$ | 38 $7\text{ L} = 7000\text{ mL} \cdot 7000 - 258 = 6742\text{ mL}$ | 50 $\frac{t}{9} \times 9 = 2 \times 9 \cdot t = 18$ ^{5.11} |
| 8 3 kg | 27 $584\text{ cm} = 5.84\text{ m} \cdot 4 + 5.84 = 9.84\text{ m}$ | 39 $121\text{ cm}, 106\text{ cm}, 112\text{ cm} \cdot 1\,060\text{ mm}, 1.12\text{ m}, 121\text{ cm}$ | 51 $5\text{ L} = 5000\text{ mL} \cdot 5000 - 163 = 4837\text{ mL}$ |
| 9 5.8 ^{7.04} | 28 $1 \times 60 = 60 \cdot 60 + 54 = 114\text{ min}$ | 40 $1\,000\text{ m} = 1\text{ km} \cdot 77 \div 1\,000 = 0.077\text{ km}$ ^{7.04} | 52 0.0083 ^{7.04} |
| 10 30 kL | 29 0.0836 ^{7.04} | 41 $8 \times 245 = 1960\text{ g} \cdot = 1.96\text{ kg}$ | 53 $4 \times 60 = 240 \cdot 240 + 42 = 282\text{ min}$ |
| 11 300 seconds | 30 kL to L: $\times 1\,000 \cdot 2.7 \times 1\,000 = 2\,700\text{ L}$ | 42 $x = 135$ ^{6.07} | 54 $117\text{ cm}, 114.7\text{ cm}, 124\text{ cm} \cdot 1\,147\text{ mm}, 1.17\text{ m}, 124\text{ cm}$ |
| 12 510 mm | 31 100 ^{7.04} | 43 $6\text{ L} = 6000\text{ mL} \cdot 6000 - 243 = 5757\text{ mL}$ | 55 100 ^{7.04} |
| 13 2 ^{6.04} | | | 56 $5 \times 171 = 855\text{ g} \cdot = 0.855\text{ kg}$ |
| 14 m to mm: $\times 1\,000 \cdot 3.1 \times 1\,000 = 3\,100\text{ mm}$ | | | 57 0.96 ^{7.02} |
| 15 16 kL | | | |
| 16 480 seconds | | | |
| 17 4.6 ^{7.04} | | | |
| 18 32 000 m | | | |
| 19 L to kL: $\div 1\,000 \cdot 24\,000 \div 1\,000 = 24\text{ kL}$ | | | |

8.02 Perimeter (booklet p. 5)

Examples (model these live)

E1 Find the perimeter of an 8 m by 5 m rectangle.
 $8 + 5 + 8 + 5 \cdot = 26\text{ m}$

E2 Find the perimeter. (Not to scale.)
missing: $12 - 7 = 5, 10 - 4 = 6 \cdot 7 + 4 + 5 + 6 + 12 + 10 = 44\text{ cm}$

E3 A square has a perimeter of 36 cm. Find a side.
4 equal sides $\cdot 36 \div 4 = 9\text{ cm}$

Questions (31 on today's skill, 10 from earlier lessons)

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|---|---|--|--|
| 1 24 m | 13 missing: 4 and $7 \cdot 8 + 6 + 4 + 7 + 12 + 13 = 50\text{ cm}$ | 23 $2 \times 27 = 54 \cdot 66 - 54 = 12 \cdot$ width: 6 m | 33 $P = 82\text{ m} \cdot 82 \times 16 = \$1\,312$ |
| 2 82 cm | 14 $8 \times 14 \cdot = 112\text{ cm}$ | 24 $8 \times 8 \cdot = 64\text{ cm}$ | 34 square: $120 \div 4 = 30\text{ cm} \cdot$ triangle: $120 \div 3 = 40\text{ cm}$ |
| 3 76 cm | 15 cm to mm: $\times 10 \cdot 56 \times 10 = 560\text{ mm}$ ^{8.01} | 25 $2 \times 16 = 32 \cdot 58 - 32 = 26 \cdot$ width: 13 m | 35 1 ^{6.03} |
| 4 42 m | 16 18 cm | 26 $P = 34\text{ m} \cdot 34 \times 16 = \544 | 36 $2 \times 30 = 60 \cdot 76 - 60 = 16 \cdot$ width: 8 m |
| 5 82 cm | 17 missing: 3 and $4 \cdot 8 + 3 + 3 + 4 + 11 + 7 = 36\text{ cm}$ | 27 $518\text{ cm} = 5.18\text{ m} \cdot 4 + 5.18 = 9.18\text{ m}$ ^{8.01} | 37 26 kg ^{8.01} |
| 6 32 cm | 18 $8 \times 4 \cdot = 32\text{ cm}$ | 28 square: $144 \div 4 = 36\text{ cm} \cdot$ triangle: $144 \div 3 = 48\text{ cm}$ | 38 20 cm |
| 7 34 m | 19 20 cm | 29 $a = 70$ ^{6.09} | 39 $P = 52\text{ m} \cdot 52 \times 10 = \520 |
| 8 54 cm | 20 23 ^{8.01} | 30 $2 \times 30 = 60 \cdot 82 - 60 = 22 \cdot$ width: 11 m | 40 $4 \times 60 = 240 \cdot 240 + 7 = 247\text{ min}$ ^{8.01} |
| 9 24 cm | 21 missing: 3 and $8 \cdot 10 + 4 + 3 + 8 + 13 + 12 = 50\text{ cm}$ | 31 $P = 72\text{ m} \cdot 72 \times 22 = \$1\,584$ | 41 square: $96 \div 4 = 24\text{ cm} \cdot$ triangle: $96 \div 3 = 32\text{ cm}$ |
| 10 26 m | 22 $1 \times 60 = 60 \cdot 60 + 46 = 106\text{ min}$ ^{8.01} | 32 180 minutes ^{8.01} | |
| 11 12 ^{7.10} | | | |
| 12 missing: 3 and $7 \cdot 6 + 3 + 3 + 7 + 9 + 10 = 38\text{ cm}$ | | | |

8.03 Circumference of a circle (booklet p. 7)

Examples (model these live)

E1 Find the circumference when $d = 10\text{ cm}$ (1 decimal place).
 $C = \pi \times 10 \cdot \approx 31.4\text{ cm}$

E2 Find the circumference when $r = 6\text{ m}$ (1 decimal place).
 $C = 2 \times \pi \times 6 \cdot \approx 37.7\text{ m}$

E3 A circle has a circumference of 50 cm. Find its diameter.
 $d = C \div \pi = 50 \div \pi \cdot \approx 15.9\text{ cm}$

Questions (33 on today's skill, 13 from earlier lessons)

- | | | | |
|-----------|--------|-----------|------------|
| 1 62.8 cm | 2 14 m | 3 75.4 cm | 4 106.8 cm |
|-----------|--------|-----------|------------|

- | | | | |
|---|---|--|---|
| 5 56 m | 18 $C = \pi \times 44 \cdot \approx 138.2 \text{ cm}$ | 30 True: it has 4 right angles and opposite sides equal. ^{6.08} | 38 27.3 ^{7.10} |
| 6 37.7 cm | 19 $862 \text{ cm} = 8.62 \text{ m} \cdot 6 + 8.62 = 14.62 \text{ m}$ ^{8.01} | 31 $d = C \div \pi \cdot = 82 \div \pi \approx 26.1 \text{ cm}$ | 39 curved part: $\pi \times 12 \div 2 \approx 18.8 \cdot + 12: \approx 30.8 \text{ cm}$ |
| 7 28.3 cm | 20 $C = 2 \times \pi \times 17 \cdot \approx 106.8 \text{ m}$ | 32 curved part: $\pi \times 19 \div 2 \approx 29.8 \cdot + 19: \approx 48.8 \text{ cm}$ | 40 one turn: $\pi \times 67 \approx 210.5 \text{ cm} \cdot \times 49 \approx 103.1 \text{ m}$ |
| 8 18 m | 21 $C = \pi \times 58 \cdot \approx 182.2 \text{ cm}$ | 33 missing: 2 and 8 $\cdot 8 + 3 + 2 + 8 + 10 + 11 = 42 \text{ cm}$ ^{8.02} | 41 $\frac{m}{4} = 9 \cdot m = 9 \times 4 = 36$ ^{5.12} |
| 9 20 m | 22 0.0487 ^{7.04} | 34 one turn: $\pi \times 53 \approx 166.5 \text{ cm} \cdot \times 49 \approx 81.6 \text{ m}$ | 42 $d = C \div \pi \cdot = 25 \div \pi \approx 8.0 \text{ cm}$ |
| 10 31.4 cm | 23 $C = \pi \times 42 \cdot \approx 131.9 \text{ cm}$ | 35 8 ^{7.08} | 43 46 cm ^{8.02} |
| 11 88.0 cm | 24 $C = 2 \times \pi \times 12 \cdot \approx 75.4 \text{ m}$ | 36 $d = C \div \pi \cdot = 35 \div \pi \approx 11.1 \text{ cm}$ | 44 $C = \pi \times 52 \cdot \approx 163.4 \text{ cm}$ |
| 12 21 ^{7.05} | 25 44 cm ^{8.02} | 37 curved part: $\pi \times 26 \div 2 \approx 40.8 \cdot + 26: \approx 66.8 \text{ cm}$ | 45 curved part: $\pi \times 22 \div 2 \approx 34.6 \cdot + 22: \approx 56.6 \text{ cm}$ |
| 13 $C = 2 \times \pi \times 3 \cdot \approx 18.8 \text{ m}$ | 26 $d = C \div \pi \cdot = 88 \div \pi \approx 28.0 \text{ cm}$ | | 46 \$8.49 ^{7.10} |
| 14 48 m | 27 $1 \div 6 = 0.1667 \dots \cdot \approx 0.17$ ^{7.10} | | |
| 15 81.7 cm | 28 $C = \pi \times 80 \cdot \approx 251.3 \text{ cm}$ | | |
| 16 46 ^{7.10} | 29 $C = \pi \times 22 \cdot \approx 69.1 \text{ m}$ | | |
| 17 $C = \pi \times 16 \cdot \approx 50.3 \text{ m}$ | | | |

8.04 Area units (booklet p. 9)

Examples (model these live)

- | | | |
|---|--|--|
| E1 Change 3 cm ² to mm ² .
$1 \text{ cm}^2 = 100 \text{ mm}^2 \cdot 3 \times 100 = 300 \text{ mm}^2$ | E2 Change 25 000 cm ² to m ² .
$\div 10\ 000 \cdot = 2.5 \text{ m}^2$ | E3 Change 4.5 ha to m ² .
$\times 10\ 000 \cdot = 45\ 000 \text{ m}^2$ |
|---|--|--|

Questions (39 on today's skill, 15 from earlier lessons)

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|--|---|---|---|
| 1 2 900 mm ² | 19 $1 \text{ m}^2 = 100 \times 100 = 10\ 000 \text{ cm}^2 \cdot 8.5 \times 10\ 000 = 85\ 000 \text{ cm}^2$ | 31 $C = 2 \times \pi \times 10 \cdot \approx 62.8 \text{ m}$ ^{8.03} | 43 $0.6 \text{ m}^2 = 6\ 000 \text{ cm}^2 \cdot$
larger: 0.6 m^2 |
| 2 5 ha | 20 $\div 100 \cdot = 89 \text{ cm}^2$ | 32 $\div 100 \cdot = 67 \text{ cm}^2$ | 44 22.0 cm ^{8.03} |
| 3 100 | 21 $8 \times 15 \cdot = 120 \text{ cm}$ ^{8.02} | 33 $1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2 \cdot 4 \times 100 = 400 \text{ mm}^2$ | 45 $51 \times 26 = 1\ 326 \text{ cm}^2 \cdot \div 10\ 000 = 0.1326 \text{ m}^2$ |
| 4 2 400 mm ² | 22 $1 \text{ m}^2 = 100 \times 100 = 10\ 000 \text{ cm}^2 \cdot 7.5 \times 10\ 000 = 75\ 000 \text{ cm}^2$ | 34 True: it has 4 right angles and opposite sides equal. ^{6.08} | 46 $0.6 \text{ m}^2 = 6\ 000 \text{ cm}^2 \cdot$
larger: $7\ 000 \text{ cm}^2$ |
| 5 19 cm ² | 23 $\div 100 \cdot = 91 \text{ cm}^2$ | 35 $A = 500 \times 500 = 250\ 000 \text{ m}^2 \cdot \div 10\ 000 = 25 \text{ ha}$ | 47 $77 \times 39 = 3\ 003 \text{ cm}^2 \cdot \div 10\ 000 = 0.3003 \text{ m}^2$ |
| 6 10 000 | 24 16 cm ^{6.10} | 36 24 cm ^{8.02} | 48 $180 - 28 = 152 \cdot 152 \div 2 = 76^\circ$ ^{6.06} |
| 7 200 000 cm ² | 25 $1 \text{ km}^2 = 1000 \times 1000 = 1\ 000\ 000 \text{ m}^2 \cdot 8 \times 1\ 000\ 000 = 8\ 000\ 000 \text{ m}^2$ | 37 $A = 200 \times 400 = 80\ 000 \text{ m}^2 \cdot \div 10\ 000 = 8 \text{ ha}$ | 49 $A = 600 \times 200 = 120\ 000 \text{ m}^2 \cdot \div 10\ 000 = 12 \text{ ha}$ |
| 8 30 cm ² | 26 $\div 100 \cdot = 87 \text{ cm}^2$ | 38 $0.8 \text{ m}^2 = 8\ 000 \text{ cm}^2 \cdot$
larger: 0.8 m^2 | 50 5 000 mL ^{8.01} |
| 9 39 ^{8.01} | 27 $\div 100 \cdot = 35 \text{ cm}^2$ | 39 cm to mm: $\times 10 \cdot 41 \times 10 = 410 \text{ mm}$ ^{8.01} | 51 $\div 100 \cdot = 63 \text{ cm}^2$ |
| 10 9 ha | 28 $1 \text{ m}^2 = 100 \times 100 = 10\ 000 \text{ cm}^2 \cdot 9.5 \times 10\ 000 = 95\ 000 \text{ cm}^2$ | 40 $47 \times 33 = 1\ 551 \text{ cm}^2 \cdot \div 10\ 000 = 0.1551 \text{ m}^2$ | 52 $0.5 \text{ m}^2 = 5\ 000 \text{ cm}^2 \cdot$
larger: $7\ 000 \text{ cm}^2$ |
| 11 10 000 | 29 300 minutes ^{8.01} | 41 $x = 115$ ^{6.07} | 53 $C = \pi \times 28 \cdot \approx 88.0 \text{ m}$ ^{8.03} |
| 12 300 000 m ² | 30 $A = 800 \times 200 = 160\ 000 \text{ m}^2 \cdot \div 10\ 000 = 16 \text{ ha}$ | 42 $A = 300 \times 200 = 60\ 000 \text{ m}^2 \cdot \div 10\ 000 = 6 \text{ ha}$ | 54 $42 \times 26 = 1\ 092 \text{ cm}^2 \cdot \div 10\ 000 = 0.1092 \text{ m}^2$ |
| 13 a full turn: back where it started ^{6.02} | | | |
| 14 $1 \text{ cm}^2 = 10 \times 10 = 100 \text{ mm}^2 \cdot 3.5 \times 100 = 350 \text{ mm}^2$ | | | |
| 15 2 m ² | | | |
| 16 140 000 m ² | | | |
| 17 10 m ^{8.03} | | | |
| 18 $1 \text{ m}^2 = 100 \times 100 = 10\ 000 \text{ cm}^2 \cdot 5 \times 10\ 000 = 50\ 000 \text{ cm}^2$ | | | |

8.05 Area of squares and rectangles (booklet p. 11)

Examples (model these live)

- | | | |
|---|--|---|
| E1 Find the area of an 8 m by 5 m rectangle.
$A = 8 \times 5 \cdot = 40 \text{ m}^2$ | E2 Find the area of a square with sides of 7 cm.
$A = 7 \times 7 \cdot = 49 \text{ cm}^2$ | E3 A rectangle has an area of 60 cm ² and a width of 5 cm. Find its length.
$60 = l \times 5 \cdot l = 60 \div 5 = 12 \text{ cm}$ |
|---|--|---|

Questions (36 on today's skill, 13 from earlier lessons)

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|-----------------------|--|--|--|
| 1 27 m ² | 9 144 cm ² | 16 $252 = l \times 14 \cdot l = 252 \div 14 = 18 \text{ m}$ | 23 $A = 8 \times 5 \cdot = 40 \text{ m}^2$ |
| 2 64 cm ² | 10 30 cm ² | 17 $A = 4 \times 7 \cdot = 28 \text{ m}^2$ | 24 $\sqrt{81} \cdot = 9 \text{ cm}$ |
| 3 247 cm ² | 11 144 m ² | 18 $466 \text{ cm} = 4.66 \text{ m} \cdot 2 + 4.66 = 6.66 \text{ m}$ ^{8.01} | 25 56 cm ^{6.10} |
| 4 16 m ² | 12 6.9 ^{7.03} | 19 $\sqrt{25} \cdot = 5 \text{ cm}$ | 26 $154 = l \times 11 \cdot l = 154 \div 11 = 14 \text{ m}$ |
| 5 36 cm ² | 13 $72 = l \times 12 \cdot l = 72 \div 12 = 6 \text{ m}$ | 20 $120 = l \times 10 \cdot l = 120 \div 10 = 12 \text{ m}$ | 27 540 seconds ^{8.01} |
| 6 100 cm ² | 14 25 cm ² | 21 $C = \pi \times 72 \cdot \approx 226.2 \text{ cm}$ ^{8.03} | 28 $A = 8 \times 4 = 32 \text{ m}^2 \cdot 32 \times 56 = \$1\ 792$ |
| 7 182 m ² | 15 26 cm ² | 22 $A = 3.4 \times 6 \cdot = 20.4 \text{ m}^2$ | 29 $A = 3.2 \times 6 \cdot = 19.2 \text{ m}^2$ |
| 8 121 cm ² | | | 30 80 cm ^{8.02} |

- 31 $\sqrt{121} = 11$ cm
- 32 $1 \text{ cm}^2 = 10 \times 10 = 100$
 $\text{mm}^2 \cdot 3 \times 100 = 300 \text{ mm}^2$ ^{8.04}
- 33 $A = 8 \times 6 = 48 \text{ m}^2 \cdot 48 \times 37 = \$1\,776$
- 34 $600 \div 20 = 30, 300 \div 20 = 15 \cdot 30 \times 15 = 450$ tiles
- 35 $0.060, 0.180, 0.900, 0.447 \cdot 0.9, 0.447, 0.18, 0.06$ ^{7.01}
- 36 big: $10 \times 10 = 100 \cdot \text{lawn} = 64 \cdot \text{path} = 36 \text{ m}^2$
- 37 125.7 cm ^{8.03}
- 38 $A = 8 \times 4 = 32 \text{ m}^2 \cdot 32 \times 39 = \$1\,248$
- 39 $800 \div 20 = 40, 300 \div 20 = 15 \cdot 40 \times 15 = 600$ tiles
- 40 mm to cm: $\div 10 \cdot 420 \div 10 = 42 \text{ cm}$ ^{8.01}
- 41 $500 \div 20 = 25, 400 \div 20 = 20 \cdot 25 \times 20 = 500$ tiles
- 42 big: $11 \times 6 = 66 \cdot \text{lawn} = 36 \cdot \text{path} = 30 \text{ m}^2$
- 43 23.8 ^{7.05}
- 44 $A = 5 \times 6 = 30 \text{ m}^2 \cdot 30 \times 51 = \$1\,530$
- 45 2.5 L ^{8.01}
- 46 $\sqrt{196} = 14$ cm
- 47 $600 \div 20 = 30, 600 \div 20 = 30 \cdot 30 \times 30 = 900$ tiles
- 48 $210\,000 \text{ m}^2$ ^{8.04}
- 49 big: $14 \times 8 = 112 \cdot \text{lawn} = 72 \cdot \text{path} = 40 \text{ m}^2$

8.06 Area of a triangle (booklet p. 13)

Examples (model these live)

- E1 Find the area: base 10 cm, height 6 cm.
 $A = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$
- E2 Find the area. (Not to scale.)
 $A = \frac{1}{2} \times 12 \times 5 = 30 \text{ m}^2$
- E3 A triangle has an area of 24 cm^2 and a base of 8 cm. Find its height.
 $24 = \frac{1}{2} \times 8 \times h = 4h \cdot h = 6 \text{ cm}$

Questions (30 on today's skill, 10 from earlier lessons)

- 1 91 cm^2
- 2 72 cm^2
- 3 16 cm^2
- 4 75 cm^2
- 5 49 cm^2
- 6 33 cm^2
- 7 19.5 cm^2
- 8 13.5 cm^2
- 9 24 cm^2
- 10 2 ^{7.08}
- 11 $A = \frac{1}{2} \times 9 \times 8 = 36 \text{ m}^2$
- 12 $A = \frac{1}{2} \times 4 \times 9 = 18 \text{ m}^2$
- 13 $A = \frac{1}{2} \times 4 \times 4 = 8 \text{ m}^2$
- 14 $154 \text{ cm} = 1.54 \text{ m} \cdot 2 + 1.54 = 3.54 \text{ m}$ ^{8.01}
- 15 base and height are the two shorter sides $\cdot \frac{1}{2} \times 6 \times 12 = 36 \text{ cm}^2$
- 16 $A = \frac{1}{2} \times 9.8 \times 16 = 78.4 \text{ m}^2$
- 17 base and height are the two shorter sides $\cdot \frac{1}{2} \times 4 \times 8 = 16 \text{ cm}^2$
- 18 $\$11.70$ ^{7.11}
- 19 $A = \frac{1}{2} \times 12 \times 11 = 66 \text{ m}^2$
- 20 56 cm ^{8.02}
- 21 $77 = \frac{1}{2} \times 11 \times h = 5.5h \cdot h = 14 \text{ cm}$
- 22 $A = \frac{1}{2} \times 5.5 \times 6 = 16.5 \text{ m}^2$
- 23 $36 = \frac{1}{2} \times 6 \times h = 3h \cdot h = 12 \text{ cm}$
- 24 Both are 30 cm^2 : area depends only on base and height.
 $25 = \frac{5}{100} = 0.05$ ^{7.02}
- 26 square: $64 \text{ cm}^2 \cdot \text{half} = 32 \text{ cm}^2$
- 27 88.0 cm ^{8.03}
- 28 $52 = \frac{1}{2} \times 8 \times h = 4h \cdot h = 13 \text{ cm}$
- 29 square: $81 \text{ cm}^2 \cdot \text{half} = 40.5 \text{ cm}^2$
- 30 $4 \times 60 = 240 \cdot 240 + 36 = 276 \text{ min}$ ^{8.01}
- 31 $152 = \frac{1}{2} \times 19 \times h = 9.5h \cdot h = 16 \text{ cm}$
- 32 square: $144 \text{ cm}^2 \cdot \text{half} = 72 \text{ cm}^2$
- 33 $66 = \frac{1}{2} \times 12 \times h = 6h \cdot h = 11 \text{ cm}$
- 34 $a = 120$ ^{6.09}
- 35 base and height are the two shorter sides $\cdot \frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2$
- 36 32 cm ^{8.02}
- 37 square: $16 \text{ cm}^2 \cdot \text{half} = 8 \text{ cm}^2$
- 38 $90 = \frac{1}{2} \times 20 \times h = 10h \cdot h = 9 \text{ cm}$
- 39 $260\,000 \text{ m}^2$ ^{8.04}
- 40 base and height are the two shorter sides $\cdot \frac{1}{2} \times 9 \times 9 = 40.5 \text{ cm}^2$

8.07 Area of a parallelogram (booklet p. 15)

Examples (model these live)

- E1 Find the area: base 9 cm, height 4 cm.
 $A = 9 \times 4 = 36 \text{ cm}^2$
- E2 Find the area. (Not to scale.)
 $A = 15 \times 6 = 90 \text{ m}^2$
- E3 A parallelogram has an area of 56 m^2 and a height of 7 m. Find its base.
 $56 = b \times 7 \cdot b = 8 \text{ m}$

Questions (27 on today's skill, 10 from earlier lessons)

- 1 55 m^2
- 2 60 cm^2
- 3 156 m^2
- 4 153 cm^2
- 5 120 m^2
- 6 24 cm^2
- 7 156 m^2
- 8 $A = 15 \times 6 = 90 \text{ m}^2$
- 9 0.58 ^{7.02}
- 10 126 cm^2
- 11 $A = 15 \times 8 = 120 \text{ m}^2$
- 12 $80 = b \times 8 \cdot b = 10 \text{ m}$
- 13 $418 \text{ cm} = 4.18 \text{ m} \cdot 4 + 4.18 = 8.18 \text{ m}$ ^{8.01}
- 14 $A = 9.7 \times 6 = 58.2 \text{ cm}^2$
- 15 $21 = b \times 3 \cdot b = 7 \text{ m}$
- 16 $A = 7.2 \times 8 = 57.6 \text{ cm}^2$
- 17 6 right ^{6.01}
- 18 $A = 14 \times 6 = 84 \text{ m}^2$
- 19 44 cm ^{8.02}
- 20 $A = 6 \times 10 = 60 \text{ cm}^2 \cdot P = 2 \times 6 + 2 \times 12 = 36 \text{ cm}$
- 21 $A = 6 \times 4 = 24 \text{ cm}^2 \cdot P = 2 \times 6 + 2 \times 6 = 24 \text{ cm}$
- 22 Cut off the triangle at one end and move it: it makes a rectangle with that height.
- 23 $15.90 \div 2 = \$7.95$ ^{7.07}
- 24 parallelogram: $48 \text{ m}^2 \cdot \text{each triangle} = 24 \text{ m}^2$
- 25 $10\,000$ ^{8.04}
- 26 $A = 11 \times 3 = 33 \text{ cm}^2 \cdot P = 2 \times 11 + 2 \times 4 = 30 \text{ cm}$
- 27 They are equal: both are base $\times$ height.
- 28 $C = \pi \times 44 \approx 138.2 \text{ cm}$ ^{8.03}
- 29 parallelogram: $35 \text{ m}^2 \cdot \text{each triangle} = 17.5 \text{ m}^2$
- 30 $A = 12 \times 10 = 120 \text{ cm}^2 \cdot P = 2 \times 12 + 2 \times 13 = 50 \text{ cm}$
- 31 equilateral ^{6.05}
- 32 $A = 3.4 \times 9 = 30.6 \text{ cm}^2$
- 33 540 minutes ^{8.01}
- 34 parallelogram: $27 \text{ m}^2 \cdot \text{each triangle} = 13.5 \text{ m}^2$
- 35 $A = 7 \times 7 = 49 \text{ cm}^2 \cdot P = 2 \times 7 + 2 \times 8 = 30 \text{ cm}$
- 36 180 m^2 ^{8.05}
- 37 $A = 6 \times 2 = 12 \text{ cm}^2$

8.08 Area of composite shapes (booklet p. 17)

Examples (model these live)

- E1 Find the area of the L-shape. (Not to scale.)
 $7 \times 10 = 70, 5 \times 6 = 30 \cdot 70 + 30 = 100 \text{ cm}^2$
- E2 A $12 \text{ cm} \times 8 \text{ cm}$ card has a $3 \text{ cm} \times 2 \text{ cm}$ hole cut out. Find the area left.
 $12 \times 8 = 96, 3 \times 2 = 6 \cdot 96 - 6 = 90 \text{ cm}^2$
- E3 Find the area of the shape. (Not to scale.)
square: $10 \times 8 = 80$, triangle: $\frac{1}{2} \times 10 \times 4 = 20 \cdot 80 + 20 = 100 \text{ m}^2$

Questions (24 on today's skill, 8 from earlier lessons)

- | | | | |
|---|--|--|--|
| 1 $54 + 16 \cdot = 70 \text{ m}^2$ | 11 $96 - 6 \cdot = 90 \text{ cm}^2$ | 17 $1 \text{ }^{6.04}$ | 26 $144 - \frac{1}{2} \times 6 \times 6 \cdot = 144 -$ |
| 2 $81 - 9 \cdot = 72 \text{ cm}^2$ | 12 $6 \times 12 = 72 \cdot 2 \times 4 = 8 \cdot$ | 18 add: $78 + 33 \cdot$ subtract: | $18 = 126 \text{ cm}^2$ |
| 3 $24 + 24 \cdot = 48 \text{ m}^2$ | total: 80 cm^2 | $117 - 6 \cdot = 111 \text{ cm}^2$ | 27 $120 - 64 \cdot = 56 \text{ m}^2$ |
| 4 $63 - 6 \cdot = 57 \text{ cm}^2$ | 13 rectangle: $12 \times 7 = 84 \cdot$ | 19 $76 \text{ cm }^{8.02}$ | 28 $52 \text{ cm }^{6.10}$ |
| 5 $44 + 42 \cdot = 86 \text{ m}^2$ | triangle: $\frac{1}{2} \times 12 \times 4 = 24 \cdot$ | 20 add: $48 + 25 \cdot$ subtract: 78 | 29 add: $56 + 96 \cdot$ subtract: |
| 6 $54 - 1 \cdot = 53 \text{ cm}^2$ | total: 108 m^2 | $- 15 \cdot = 73 \text{ cm}^2$ | $224 - 72 \cdot = 152 \text{ cm}^2$ |
| 7 $44 - 3 \cdot = 41 \text{ cm}^2$ | 14 $768 \text{ cm} = 7.68 \text{ m} \cdot 2 + 7.68 =$ | 21 $121 - \frac{1}{2} \times 4 \times 4 \cdot = 121 - 8$ | 30 $30 \text{ m}^2 \text{ }^{8.04}$ |
| 8 $24 + 63 \cdot = 87 \text{ m}^2$ | $9.68 \text{ m }^{8.01}$ | $= 113 \text{ cm}^2$ | 31 rectangle: $18 \times 9 = 162 \cdot$ |
| 9 $6 \times 9 = 54 \cdot 2 \times 6 = 12 \cdot$ | 15 rectangle: $18 \times 9 = 162 \cdot$ | 22 $\times 100: 100 \div 4 \cdot = 25 \text{ }^{7.08}$ | triangle: $\frac{1}{2} \times 18 \times 4 = 36 \cdot$ total: |
| total: 66 cm^2 | triangle: $\frac{1}{2} \times 18 \times 6 = 54 \cdot$ total: | 23 $140 - 72 \cdot = 68 \text{ m}^2$ | 198 m^2 |
| 10 $16 \text{ }^{5.13}$ | 216 m^2 | 24 $100 \text{ m}^2 \text{ }^{8.07}$ | 32 $64 - \frac{1}{2} \times 4 \times 4 \cdot = 64 - 8 =$ |
| | 16 $10 \times 12 = 120 \cdot 6 \times 10 = 60$ | 25 add: $84 + 20 \cdot$ subtract: | 56 cm^2 |
| | $\cdot$ total: 180 cm^2 | $112 - 8 \cdot = 104 \text{ cm}^2$ | |

8.09 Prisms, cross-sections and views (booklet p. 19)

Examples (model these live)

- | | | |
|---|--|--|
| E1 Name the prism.
the cross-section is a triangle
triangular prism | E2 How many faces, edges and
vertices does a rectangular prism have?
6 faces, 12 edges
8 vertices | E3 Is a square pyramid a prism?
Explain.
its cross-section changes
no: it
narrows to a point |
|---|--|--|

Questions (34 on today's skill, 14 from earlier lessons)

- | | | | |
|--|---|---|---|
| 1 rectangular prism | 16 no | 27 $19 \text{ }^{5.13}$ | 38 a rectangle |
| 2 5 | 17 $108 - 2 \cdot = 106 \text{ cm}^2 \text{ }^{8.08}$ | 28 faces: $5 + 2 = 7$, edges: $3 \times$ | 39 $11 - 2 = 9$ sides
a 9-sided |
| 3 no | 18 faces: $8 + 2 = 10$, edges: 3 | $5 = 15 \cdot$ vertices: $2 \times 5 = 10$ | polygon; 27 edges |
| 4 pentagonal prism | $\times 8 = 24 \cdot$ vertices: $2 \times 8 = 16$ | 29 $44 \text{ cm }^{8.02}$ | 40 $C = \pi \times 56 \cdot \approx 175.9 \text{ cm }^{8.03}$ |
| 5 8 | 19 $F = 6$, $V = 8$, $E = 12 \cdot 6 + 8 -$ | 30 a square with its two | 41 $10 - 2 = 8$ sides
a 8-sided |
| 6 no | $12 = 2 \checkmark$ | diagonals | polygon; 24 edges |
| 7 hexagonal prism | 20 $396 \text{ cm} = 3.96 \text{ m} \cdot 8 + 3.96$ | 31 $F = 7$, $V = 10$, $E = 15 \cdot 7 +$ | 42 a rectangle (with a line |
| 8 6 | $= 11.96 \text{ m }^{8.01}$ | $10 - 15 = 2 \checkmark$ | along the middle) |
| 9 square $^{6.08}$ | 21 a rectangle 1 cm wide | 32 $50 \text{ cm}^2 \text{ }^{8.06}$ | 43 $1 \text{ }^{6.03}$ |
| 10 7 | and 2 cm high | 33 a triangle | 44 $12 - 2 = 10$ sides
a 10- |
| 11 no (it has a curved | 22 faces: $3 + 2 = 5$, edges: $3 \times$ | 34 $13 - 2 = 11$ sides
a 11- | sided polygon; 30 edges |
| surface) | $3 = 9 \cdot$ vertices: $2 \times 3 = 6$ | sided polygon; 33 edges | 45 $17 \text{ m}^2 \text{ }^{8.04}$ |
| 12 triangular prism | 23 $70 = b \times 5 \cdot b = 14 \text{ m }^{8.07}$ | 35 obtuse-angled
scalene | 46 $9 - 2 = 7$ sides
a 7-sided |
| 13 $a = 70 \text{ }^{6.06}$ | 24 $F = 5$, $V = 6$, $E = 9 \cdot 5 + 6 -$ | (no equal angles) $^{6.05}$ | polygon; 21 edges |
| 14 faces: $6 + 2 = 8$, edges: $3 \times$ | $9 = 2 \checkmark$ | 36 hexagonal prism ($6 \times 2 =$ | 47 $14 - 2 = 12$ sides
a 12- |
| $6 = 18 \cdot$ vertices: $2 \times 6 = 12$ | 25 $F = 8$, $V = 12$, $E = 18 \cdot 8 +$ | $12)$ | sided polygon; 36 edges |
| 15 6 | $12 - 18 = 2 \checkmark$ | 37 $36 + 14 \cdot = 50 \text{ m}^2 \text{ }^{8.08}$ | 48 $45 \text{ cm}^2 \text{ }^{8.06}$ |
| | 26 a rectangle 3 cm by 1 cm | | |

8.10 Volume units (booklet p. 21)

Examples (model these live)

- | | | |
|--|---|--|
| E1 Change 5 cm^3 to mm^3 .
$\times 1000 \cdot = 5000 \text{ mm}^3$ | E2 Change 2.5 m^3 to cm^3 .
$\times 1000000 \cdot = 2500000 \text{ cm}^3$ | E3 How many litres does 3 m^3 hold?
$1 \text{ m}^3 = 1000 \text{ L} \cdot 3000 \text{ L}$ |
|--|---|--|

Questions (40 on today's skill, 16 from earlier lessons)

- | | | | |
|--|--|--|---|
| 1 11000 mm^3 | 15 755 mL | 25 $1 \text{ m}^3 = 100 \times 100 \times 100 = 1$ | 33 $1 \text{ m}^3 = 100 \times 100 \times 100 = 1$ |
| 2 126 mL | 16 3000 L | $000000 \text{ cm}^3 \cdot = 850000$ | $000000 \text{ cm}^3 \cdot = 700000$ |
| 3 4000 L | 17 $25000 \text{ kg }^{8.01}$ | cm^3 | cm^3 |
| 4 9000 mm^3 | 18 3000 mm^3 | 26 $2500 \text{ cm}^3 = 2500 \text{ mL} \cdot =$ | 34 $80 - 9 \cdot = 71 \text{ cm}^2 \text{ }^{8.08}$ |
| 5 743 mL | 19 $\div 1000 \cdot = 6 \text{ cm}^3$ | 2.5 L | 35 $2300 \text{ cm}^3 = 2300 \text{ mL} \cdot =$ |
| 6 2000 L | 20 $1 \text{ m}^3 = 100 \times 100 \times 100 = 1$ | 27 $1 \text{ m}^3 = 100 \times 100 \times 100 = 1$ | 2.3 L |
| 7 16000 mm^3 | $000000 \text{ cm}^3 \cdot = 4500000$ | $000000 \text{ cm}^3 \cdot = 950000$ | 36 $154 = b \times 14 \cdot b = 11 \text{ m }^{8.07}$ |
| 8 565 mL | cm^3 | cm^3 | 37 $\div 1000 \cdot = 50 \text{ cm}^3$ |
| 9 $20 \text{ m}^2 \text{ }^{8.04}$ | 21 $8 \times 7 \cdot = 56 \text{ cm }^{8.02}$ | 28 $200 \text{ cm}^3 = 200 \text{ mL} \cdot = 0.2 \text{ L}$ | 38 $0.9 \text{ m}^3 = 900000 \text{ cm}^3 \cdot$ |
| 10 153 mL | 22 $2000 \text{ cm}^3 = 2000 \text{ mL} \cdot =$ | 29 $54 \text{ }^{7.10}$ | larger: 0.9 m^3 |
| 11 9000 L | 2 L | 30 $\div 1000 \cdot = 13 \text{ cm}^3$ | 39 $20 \times 20 \times 20 = 8000 \text{ cm}^3 \cdot$ |
| 12 29000 mm^3 | 23 $\div 1000 \cdot = 43 \text{ cm}^3$ | 31 $6 \text{ m }^{8.03}$ | $= 8 \text{ L}$ |
| 13 $a = 105 \text{ }^{6.09}$ | 24 $F = 8$, $V = 12$, $E = 18 \cdot 8 +$ | 32 $0.9 \text{ m}^3 = 900000 \text{ cm}^3 \cdot$ | 40 obtuse-angled
scalene |
| 14 $\div 1000 \cdot = 18 \text{ cm}^3$ | $12 - 18 = 2 \checkmark \text{ }^{8.09}$ | larger: 0.9 m^3 | (no equal angles) $^{6.05}$ |

- 41 3 600 L
 42 100 ^{8.04}
 43 $0.6 \text{ m}^3 = 600\,000 \text{ cm}^3$ ·
 larger: 0.6 m^3
 44 $40 \times 40 \times 40 = 64\,000$
 $\text{cm}^3 \cdot = 64 \text{ L}$
 45 $A = 7.5 \times 9 \cdot = 67.5 \text{ m}^2$ ^{8.05}
 46 5 300 L
 47 $3 \times 60 = 180 \cdot 180 + 6 = 186$
 min ^{8.01}
 48 $30 \times 30 \times 30 = 27\,000 \text{ cm}^3$
 $\cdot = 27 \text{ L}$
 49 5 200 L
 50 0.18 ^{7.06}
 51 $0.6 \text{ m}^3 = 600\,000 \text{ cm}^3$ ·
 larger: 0.6 m^3
 52 39 cm^2 ^{8.06}
 53 $700 \text{ cm}^3 = 700 \text{ mL} \cdot = 0.7 \text{ L}$
 54 $60 \times 60 \times 60 = 216\,000$
 $\text{cm}^3 \cdot = 216 \text{ L}$
 55 no ^{8.09}
 56 2 700 L

8.11 Volume of a rectangular prism (booklet p. 23)

Examples (model these live)

E1 Find the volume of a box 5 cm long, 4 cm wide and 3 cm high.

$$V = 5 \times 4 \times 3 \cdot = 60 \text{ cm}^3$$

E2 Find the volume of a cube with sides of 4 cm.

$$V = 4 \times 4 \times 4 \cdot = 64 \text{ cm}^3$$

E3 A box has a volume of 120 cm^3 . It is 6 cm long and 5 cm wide. Find its height.

$$120 = 6 \times 5 \times h = 30h \cdot h = 4 \text{ cm}$$

Questions (31 on today's skill, 11 from earlier lessons)

- 1 324 cm^3
 2 200 m^3
 3 343 cm^3
 4 72 cm^3
 5 45 m^3
 6 512 cm^3
 7 154 cm^3
 8 120 m^3
 9 1000 cm^3
 10 180 cm^3
 11 $a = 75$ ^{6.06}
 12 $V = 4 \times 4 \times 8 \cdot = 128 \text{ cm}^3$
 13 $V = 4 \times 3 \times 5 \cdot = 60 \text{ cm}^3$
 14 $110 = 11 \times 5 \times h = 55h \cdot h = 2 \text{ cm}$
 15 $915 \text{ cm} = 9.15 \text{ m} \cdot 9 + 9.15 = 18.15 \text{ m}$ ^{8.01}
 16 $V = 2.3 \times 2 \times 4 \cdot = 18.4 \text{ m}^3$
 17 $V = 4 \times 3 \times 3 \cdot = 36 \text{ cm}^3$
 18 $60 = 6 \times 2 \times h = 12h \cdot h = 5 \text{ cm}$
 19 $V = 2.3 \times 4 \times 3 \cdot = 27.6 \text{ m}^3$
 20 $x = 115$ ^{6.07}
 21 $V = 10 \times 3 \times 7 \cdot = 210 \text{ cm}^3$
 22 78 cm ^{8.02}
 23 $V = 48\,000 \text{ cm}^3 \cdot = 48 \text{ L}$
 24 $27 = 3 \times 3 \times h = 9h \cdot h = 3 \text{ cm}$
 25 $V = 24\,000 \text{ cm}^3 \cdot = 24 \text{ L}$
 26 $3 \times 4 \times 4 \cdot = 48 \text{ cubes}$
 27 $\frac{y}{2} = 7 \cdot y = 7 \times 2 = 14$ ^{5.12}
 28 $64 \rightarrow 512 \text{ cm}^3 \cdot 8 \text{ times larger}$
 29 32 cm^2 ^{8.05}
 30 $V = 84\,000 \text{ cm}^3 \cdot = 84 \text{ L}$
 31 $5 \times 4 \times 2 \cdot = 40 \text{ cubes}$
 32 $C = \pi \times 59 \cdot \approx 185.4 \text{ cm}$ ^{8.03}
 33 $4 \times 4 \times 4 \cdot = 64 \text{ cubes}$
 34 $27 \rightarrow 216 \text{ cm}^3 \cdot 8 \text{ times larger}$
 35 32 cm ^{6.10}
 36 $V = 36\,000 \text{ cm}^3 \cdot = 36 \text{ L}$
 37 25 m^2 ^{8.04}
 38 $V = 2.3 \times 4 \times 4 \cdot = 36.8 \text{ m}^3$
 39 $4 \times 5 \times 3 \cdot = 60 \text{ cubes}$
 40 $20 + 50 \cdot = 70 \text{ m}^2$ ^{8.08}
 41 $8 \rightarrow 64 \text{ cm}^3 \cdot 8 \text{ times larger}$
 42 $\div 1000 \cdot = 90 \text{ cm}^3$ ^{8.10}

8.12 Volume of a prism (booklet p. 25)

Examples (model these live)

E1 A prism has a cross-section of 15 cm^2 and a length of 8 cm. Find its volume.

$$V = 15 \times 8 \cdot = 120 \text{ cm}^3$$

E2 A triangular prism has a triangle of base 6 cm and height 4 cm, and is 10 cm long. Find its volume.

$$A = \frac{1}{2} \times 6 \times 4 = 12 \cdot V = 12 \times 10 = 120 \text{ cm}^3$$

E3 A prism has a volume of 96 m^3 and a cross-section of 12 m^2 . How long is it?

$$96 = 12 \times h \cdot h = 8 \text{ m}$$

Questions (28 on today's skill, 9 from earlier lessons)

- 1 192 cm^3
 2 12 cm^2
 3 216 cm^3
 4 40 cm^2
 5 210 cm^3
 6 16 cm^2
 7 286 cm^3
 8 10 cm^2
 9 9 cm^2
 10 320 cm^3
 11 $A = \frac{1}{2} \times 12 \times 3 = 18 \cdot V = 18 \times 11 = 198 \text{ cm}^3$
 12 $x = 140$ ^{6.07}
 13 8 cm^2
 14 $A = \frac{1}{2} \times 6 \times 4 = 12 \cdot V = 12 \times 10 = 120 \text{ cm}^3$
 15 $378 = 27 \times h \cdot h = 14 \text{ m}$
 16 $490 \text{ cm} = 4.9 \text{ m} \cdot 5 + 4.9 = 9.9 \text{ m}$ ^{8.01}
 17 $A = \frac{1}{2} \times 12 \times 6 = 36 \cdot V = 36 \times 8 = 288 \text{ cm}^3$
 18 $91 = 7 \times h \cdot h = 13 \text{ m}$
 19 $A = \frac{1}{2} \times 12 \times 4 = 24 \cdot V = 24 \times 13 = 312 \text{ cm}^3$
 20 $a = 95$ ^{6.06}
 21 $A = 120 - 15 = 105 \text{ cm}^2 \cdot V = 105 \times 6 = 630 \text{ cm}^3$
 22 48 cm ^{8.02}
 23 $55 = 5 \times h \cdot h = 11 \text{ m}$
 24 $A = 110 - 18 = 92 \text{ cm}^2 \cdot V = 92 \times 3 = 276 \text{ cm}^3$
 25 $A = \frac{1}{2} \times 3 \times 2 = 3 \text{ m}^2 \cdot V = 3 \times 4 = 12 \text{ m}^3$
 26 $8 = 80 \div 10 \cdot = 2$ ^{7.05}
 27 $45 \text{ and } 135 \text{ cm}^3 \cdot \text{the longer one has 3 times the volume}$
 28 27 cm^2 ^{8.06}
 29 $A = 55 - 4 = 51 \text{ cm}^2 \cdot V = 51 \times 5 = 255 \text{ cm}^3$
 30 $A = \frac{1}{2} \times 4 \times 2 = 4 \text{ m}^2 \cdot V = 4 \times 3 = 12 \text{ m}^3$
 31 $72 \text{ and } 216 \text{ cm}^3 \cdot \text{the longer one has 3 times the volume}$
 32 0.1 ^{7.09}
 33 $A = 56 - 15 = 41 \text{ cm}^2 \cdot V = 41 \times 10 = 410 \text{ cm}^3$
 34 13 ha ^{8.04}
 35 $112 = 16 \times h \cdot h = 7 \text{ m}$
 36 $A = \frac{1}{2} \times 4 \times 3 = 6 \text{ m}^2 \cdot V = 6 \times 5 = 30 \text{ m}^3$
 37 no ^{8.09}

8.13 Capacity (booklet p. 27)

Examples (model these live)

E1 Change 2.75 L to millilitres.

$$\times 1000 \cdot = 2\,750 \text{ mL}$$

E2 A box is 20 cm by 10 cm by 15 cm. How many litres does it hold?

$$V = 3\,000 \text{ cm}^3 = 3\,000 \text{ mL} \cdot = 3 \text{ L}$$

E3 A 2 000 L tank fills at 25 L a minute. How long does it take to fill?

$$2\,000 \div 25 \cdot = 80 \text{ minutes}$$

Questions (36 on today's skill, 15 from earlier lessons)

- 1 4 300 mL
 2 4.398 L
 3 7 mL
 4 2 700 mL
 5 3.539 L
 6 449 mL
 7 8 100 mL
 8 1.538 L

- 9 84 m^3 ^{8.11}
- 10 0.581 L
- 11 657 mL
- 12 $4\,400 \text{ mL}$
- 13 $x = 135$ ^{6.07}
- 14 $V = 7\,500 \text{ cm}^3 = 7\,500 \text{ mL}$
 $\cdot = 7.5 \text{ L}$
- 15 0.218 L
- 16 683 mL
- 17 133 mL ^{8.10}
- 18 $7\,400 \text{ mL}$
- 19 $V = 5\,000 \text{ cm}^3 = 5\,000 \text{ mL}$
 $\cdot = 5 \text{ L}$
- 20 $6 \text{ L} = 6\,000 \text{ mL} \cdot 6\,000 \div 500 = 12$
- 21 $574 \text{ cm} = 5.74 \text{ m} \cdot 9 + 5.74 = 14.74 \text{ m}$ ^{8.01}
- 22 $V = 2\,000 \text{ cm}^3 = 2\,000 \text{ mL}$
 $\cdot = 2 \text{ L}$
- 23 $8 \text{ L} = 8\,000 \text{ mL} \cdot 8\,000 \div 250 = 32$
- 24 $68 = 17 \times h \cdot h = 4 \text{ m}$ ^{8.12}
- 25 $5 \text{ L} = 5\,000 \text{ mL} \cdot 5\,000 \div 250 = 20$
- 26 $V = 4\,500 \text{ cm}^3 = 4\,500 \text{ mL}$
 $\cdot = 4.5 \text{ L}$
- 27 $0.\dot{3}$ ^{7.09}
- 28 $1\,200 \div 40 \cdot = 30 \text{ minutes}$
- 29 30 cm ^{8.02}
- 30 $4 \text{ L} = 4\,000 \text{ mL} \cdot 4\,000 \div 500 = 8$
- 31 $V = 3\,000 \text{ cm}^3 = 3\,000 \text{ mL}$
 $\cdot = 3 \text{ L}$
- 32 125 cm^3 ^{8.11}
- 33 $1\,900 \div 10 \cdot = 190 \text{ minutes}$
- 34 $700 \div 20 \cdot = 35 \text{ minutes}$
- 35 $V = 6 \text{ m}^3 \cdot = 6\,000 \text{ L}$
- 36 total: $\$17.15 \cdot$ change: $\$32.85$ ^{7.11}
- 37 $50 \times 20 \times 3 = 3\,000 \text{ cm}^3 \cdot = 3 \text{ L}$
- 38 476 cm^3 ^{8.12}
- 39 $1\,800 \div 15 \cdot = 120 \text{ minutes}$
- 40 $V = 3 \text{ m}^3 \cdot = 3\,000 \text{ L}$
- 41 $C = \pi \times 75 \cdot \approx 235.6 \text{ cm}$ ^{8.03}
- 42 $V = 30 \text{ m}^3 \cdot = 30\,000 \text{ L}$
- 43 $60 \times 40 \times 4 = 9\,600 \text{ cm}^3 \cdot = 9.6 \text{ L}$
- 44 2 ^{7.08}
- 45 $1\,500 \div 10 \cdot = 150 \text{ minutes}$
- 46 8 m^2 ^{8.04}
- 47 $9 \text{ L} = 9\,000 \text{ mL} \cdot 9\,000 \div 500 = 18$
- 48 $V = 8 \text{ m}^3 \cdot = 8\,000 \text{ L}$
- 49 $14\,000 \text{ mm}^3$ ^{8.10}
- 50 $50 \times 30 \times 2 = 3\,000 \text{ cm}^3 \cdot = 3 \text{ L}$
- 51 $104 = 26 \times h \cdot h = 4 \text{ m}$ ^{8.12}